

23. (a)

X	1	2	3
P	$\frac{1}{9}$	$\frac{2}{3}$	$\frac{2}{9}$

Y	0	1	2	3
P	$\frac{8}{27}$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{1}{27}$

(b) 不独立

(c)

$Y X=1$	0	1	2	3
P	$\frac{2}{3}$	0	0	$\frac{1}{3}$

$X Y=0$	1	2	3
P	$\frac{1}{4}$	$\frac{3}{4}$	0

(d) $P(X=3|Y=2)=0$ $P(Y=2|X=3)=0$

25. $D(XY) = E(X^2Y^2) - E(XY)^2$

$$= E(X^2)E(Y^2) - E(X)^2E(Y)^2$$

$$= (D(X) + E(X)^2)(D(Y) + E(Y)^2) - E(X)^2E(Y)^2$$

$$= DXDY + E(X)^2DY + E(Y)^2DX$$

26. 设 ξ_i 分布期望为 μ , 方差为 σ^2

则 $EX = EY = n\mu$

$$EXY = E\left(\sum_{k=1}^n \xi_k\right)\left(\sum_{k=1}^n \xi_{m+k}\right) = E\left(\sum_{\substack{1 \leq i \neq j \leq n \\ m+1 \leq j \leq m+n}} \xi_i \xi_j + \sum_{i=m+1}^n \xi_i^2\right)$$

$$= (n^2 - n + m)E(\xi_1 \xi_2) + (n-m)E(\xi_1^2)$$

$$= (n^2 - n + m)\mu^2 + (n-m)(\sigma^2 + \mu^2)$$

$$= n^2\mu^2 + (n-m)\sigma^2$$

$$\sqrt{DX} = \sqrt{DY} = \sqrt{n}\sigma$$

$$\therefore r_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{DX}\sqrt{DY}} = \frac{n-m}{n}$$

27. 记 $X_i = \begin{cases} 1 & \text{第 } i \text{ 次为 } 1 \\ 0 & \text{第 } i \text{ 次不为 } 1 \end{cases}$ $Y_i = \begin{cases} 1 & \text{第 } i \text{ 次为 } 6 \\ 0 & \text{第 } i \text{ 次不为 } 6 \end{cases}$

$$\begin{aligned} \therefore E(XY) &= E\left(\sum_{i=1}^n X_i \sum_{j=1}^n Y_j\right) = \sum_{i=1}^n E(X_i Y_i) + 2 \sum_{1 \leq i < j \leq n} E(X_i Y_j) \\ &= \frac{n(n+1)}{36} \end{aligned}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = -\frac{n}{36}$$

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{DX}\sqrt{DY}} = -\frac{1}{5}$$

30. $E(X|Y)$

	$\frac{13}{7}$	$\frac{28}{15}$	$\frac{11}{5}$
P	$\frac{1}{27}$	$\frac{4}{27} \cdot \frac{5}{9}$	$\frac{5}{27}$

$$EX = \sum P E(X|Y) = \frac{52}{27}$$

31. Y_2

Y_3	-2	0	2
-3	q^3	0	0
-1	$q^2 p$	$2/9 q^2$	0
1	0	$2p^2 q$	$p^2 q$
3	0	0	$10p^3$

$\therefore E(X|Y_2)$

$E(Y_3 Y_2)$	$-2+p-q$	$p-q$	$2+p-q$
P	q^2	$2pq$	p^2

$$E(Y_{n+1} = Y_n + 1 | Y_n) = p$$

$$E(Y_{n+1} = Y_n - 1 | Y_n) = q$$

$$\therefore E(Y_{n+1} | Y_n) = Y_n + p - q$$

证: 1.

设简单随机游动的步长为 X

$$\text{则 } E(X) = \sum_{i=1}^N P(X=i) \cdot i = \frac{1}{N} \sum_{i=1}^N P(X=i) i = \frac{1}{N} E(X) = \frac{1}{N}$$

$$\begin{aligned} 2. \text{ Cov}(X - (\hat{a}Y + \hat{b}), Y) &= E((X - (\hat{a}Y + \hat{b}))Y) - E(X - (\hat{a}Y + \hat{b}))EY \\ &= E(X - (\hat{a}Y + \hat{b})) = 0 \end{aligned}$$

$$E((X - (\hat{a}Y + \hat{b}))Y) = E(XY) - \hat{a}EY^2 - \hat{b}EY$$

$$= E(XY) - \hat{a}(DY + (EY)^2) - \hat{b}EY$$

$$= E(XY) - \text{Cov}(X, Y) - \hat{a}(EY)^2 - \hat{b}EY$$

$$= E(XEY) - (\hat{a}EY + \hat{b})EY = 0$$

$$\therefore \text{Cov}(X - (\hat{a}Y + \hat{b}), Y) = 0$$